JO JOURNAL OF OSSEOINTEGRATION periodontics and prosthodontics

Sealing ability of a bioceramic sealer used in combination with cold and warm obturation techniques

D. I. K. PONTORIERO[1], G. MADARO[2], V. VANAGOLLI[2], S. BENEDICENTI[2], G. VERNIANI[3], E. FERRARI CAGIDIACO[3], S. GRANDINI[4], M. FERRARI[3]

[1] Department of Prosthodontics, University of Siena - Department of Endodontics, University of Genoa, Italy
[2] Department of Endodontics, University of Genoa, Italy
[3] Department of Prosthodontics, University of Siena, Italy
[4] Department of Endodontics, University of Siena, Italy

TO CITE THIS ARTICLE
Pontoriero DIK, Madaro G, Vanagolli V, Benedicenti S, Verniani G, Ferrari Cagidiaco E, Grandini S, Ferrari M. Sealing ability of a bioceramic sealer used in combination with cold and warm obturation techniques. J Osseointegr 2021;13(4):248-255.

DOI 10.23805 /JO.2021.13.04.11

KEYWORDS Endodontic cements, Bioceramics, Obturation techniques, Warm gutta-percha techniques, Cold gutta-percha techniques.

INTRODUCTION

Root canal obturation aims to fill and seal, in all the three dimensions, the endodontic space, in order to prevent penetration and growth of microorganisms within it, as well as percolation of tissue fluids susceptible to degradation (1,2). To achieve a proper obturation of the root canal, two main factors are crucial: the root canal obturation technique and the endodontic cement.

Several root canal obturation techniques are available, and those most commonly used can be divided into two groups.

1. Warm gutta-percha techniques, such as carrier-based obturation (3,4), continuous-wave compaction technique, plasticized technique, warm vertical compaction (5-9).
2. Cold techniques, namely lateral compaction, and single cone technique (10-13).

The purpose of the cement (14,15) is to facilitate the flow of the gutta-percha, to fill the spaces not filled by the gutta-percha itself, and to create an adhesive interface on the canal walls. However, the cement appears to be a locus minoris resistentiae for bacterial infiltration, since all types of cement currently available on the market are slightly resorbable, with the formation of empty spaces; consequently, they should be used in the smallest possible quantity.

Bioceramics are non-metallic, biocompatible, and inorganic materials that include zirconia and alumina, bioactive glass, coatings and composites, hydroxyapatite, and resorbable calcium phosphates (16). Today, new bioceramic cements can be used as substitutes of MTA or as root canal sealers (16).

BioRoot RCS (Septodont) has been reported to induce, *in vitro*, the production of osteogenic and angiogenic growth factors by human periodontal ligament cells (17). Moreover, it has lower cytotoxicity than other conventional root canal sealers, and it may induce hard tissue deposition (18,19). This material also has antimicrobial activity (20). The powder contains tricalcium silicate, zirconium oxide, and povidone; the liquid is an aqueous solution of polycarboxylate and calcium chloride. Bioceramic cements are inductive materials, meaning that during hardening, when they come in contact with tissue fluids, calcium hydroxide reacts with phosphatase enzymes, thus inducing the formation of hydroxyapatite (21).

It was also demonstrated that these types of cement could be used as either root filling materials or root canal sealers in combination with gutta-percha for filling root canals. *In vivo* studies show excellent results, although Torabinejad emphasizes that the new bio-ceramic cements are indicated by recent clinical research as the future of root canal obturation, but they still need further investigation, as there is no long-term *in vivo* and *in vitro* studies in the literature (21).

There is not a standard method used to measure the bonding of a sealer to the root dentin; for this reason, the quality of the adhesion of the root filling material is commonly tested using bond strength tests and microleakage (22). Microleakage is described as the "spread of bacteria, oral fluids, ions, and molecules into the tooth and filling material interface" (23).

Apical leakage is regarded as the most common cause of endodontic failure and is influenced by several variables, such as different obturation techniques, chemical and physical properties of the root canal filling materials, and the presence or absence of the smear layer. This is referred to as primary failure or primary infiltration.

In a recent study (24) micro-CT scans were used to compare the 3D filling performance and the presence of radiographic translucencies of three different root canal filling techniques: warm vertical condensation, carrier-based technique, and single cone with bioceramic sealant; obturation volume, pore-volume, and pore rate were investigated. No obturation technique revealed completely filled root canal systems. The filling techniques employed in this study have similar features: they were biocompatible, radiopaque, and inert, providing a stable apical and intracanal seal.

Both continuous-wave and carrier-based techniques are built on the concept of minimum sealant interface, whereas the single-cone technique of bioceramic sealant originates from Grossman's concept of the maximum interface of the sealant with the gutta-percha cone intended as support (15).

Still, few studies are available concerning this category of cements, and most of them focus on the analysis of their excellent biological properties. For this reason, in this study, it was decided to focus on their behavior within the root canal when used in association with different obturation techniques.

The aim of this study was to evaluate the sealing ability of a bioceramic endodontic sealer (BioRoot RCS) and a ZOE sealer (Pulp Canal Sealer) at the apical thirds when used in combination with three different obturation techniques.

The null hypothesis tested was as follows.

1. The bioceramic endodontic cement can seal the apex similarly to traditional endodontic ZOE cement.
2. The bioceramic endodontic cement can similarly seal the apex when used in combination either with cold or warm obturation technique.

MATERIALS AND METHODS

Eigthy intact human single-rooted maxillary premolars, recently extracted for orthodontic reasons, were included in the study. After the cleaning of remaining tissues, a ×4.5 stereomicroscope (Nikon SMZ645; Nikon, Tokyo, Japan) was employed to rule out any external radicular cracks.

Samples were stored in 0.9% saline solution (SALF SPA, Cenate Sotto, Italy) at a temperature of 37°C to prevent dehydration. After preparing the access cavity, pre-curved stainless steel manual K-files # 8, and # 10 (Sweden & Martina, Due Carrare, Italy) were used to define the working length.

All chemo-mechanical procedures for preparation of the samples were performed by two calibrated operators (GM and VV).

NiTi Mtwo File (Sweden & Martina, Due Carrare, Italy) instruments were used for root canal shaping following the manufacturer's instructions with an endodontic motor (X-SMART TM Plus; Dentsply Maillefer, Ballaigues, Switzerland) set at 250rpm; 2.5 ml of 5% sodium hypochlorite (Niclor 5 dental; Ogna Lab, Muggiò, Italy) was used after each rotary instrument by syringe-needle irrigation with a 31-gauge needle tip. Sodium hypochlorite was preheated at 50°C. Apical patency was maintained by using a # 10 K-type file after each larger file. The final irrigation phase was carried out with 2 ml of 17% EDTA (Ogna Lab, Muggiò, Italy) for 2 minutes, followed by the final rinse of 5 ml of 5% NaOCl for 5 minutes, to optimize the removal of inorganic and organic components. After instrumentation, canals were dried with paper points.

Teeth were randomly divided into five groups and filled with different techniques and materials.

- Group 1 was filled using Thermafil® Obturators (Dentsply/Maillefer) with Pulp Canal Sealer™ (Kerr), n=20.
- Group 2 was filled using Thermafil® Obturators (Dentsply/Maillefer) with BioRoot™ RCS (Septodont) bioceramic sealer, n=20.
- Group 3 was filled using warm vertical condensation with traditional Pulp Canal Sealer™ (Kerr), n=20.
- Group 4 was filled using a Single GP cone with BioRoot™ RCS (Septodont) bioceramic sealer, n=20.
- Group 5 was filled with warm vertical condensation with BioRoot™ RCS bioceramic sealer (Septodont), n=20.

After root canal obturation, all dental surfaces were covered with nail varnish, except 2 mm around the area of the tooth apex which were left exposed. A solution

of ammoniacal silver nitrate (ammoniacal silver nitrate and distilled water 1:4) was prepared, and the diluted solution was filtered with a millipore filter (0.22-nm filter, Carrigtwohill, County Cork, Ireland) mounted on a syringe. Under laboratory light, each tooth was placed in a test tube with diluted ammoniacal silver nitrate solution. After 24 h, specimens were thrice rinsed in water for 10 min. Nail varnish around the tooth was removed with acetone, and each tooth was placed in a test tube with a diluted solution of photo-developer (Kodak, Rochester, NY, USA) (ratio of 1:10 photo-developer:distilled water). After 8 hours, teeth were rinsed thrice in water for 10 min. Each tooth was embedded in transparent self-curing acrylic resin. The teeth were then sliced with a low-speed diamond saw under water cooling to prevent frictional heat (Isomet; Buehler, Lake Bluff, NY, USA) into two slices along their long axis and perpendicularly to the root.
Samples were examined with a digital microscope. Two observers (DP, MF) independently scored the amount of tracer along the interface. The apical leakage was evaluated following the procedure described by Limkangwalmongkol et al. (25). In case of a score discrepancy, the sample was evaluated again by the two operators, and a common decision was taken.
The Bonferroni and Anova tests were used to assess

sealer), which is 0.59 mm (Fig. 1). Group 2 (same sealer applied with Thermafil technique) reached an average of

FIG. 1 The average of apical leakage was found in Group 4.

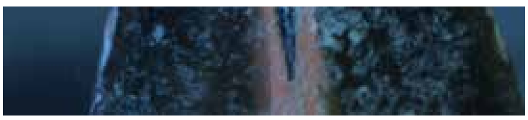
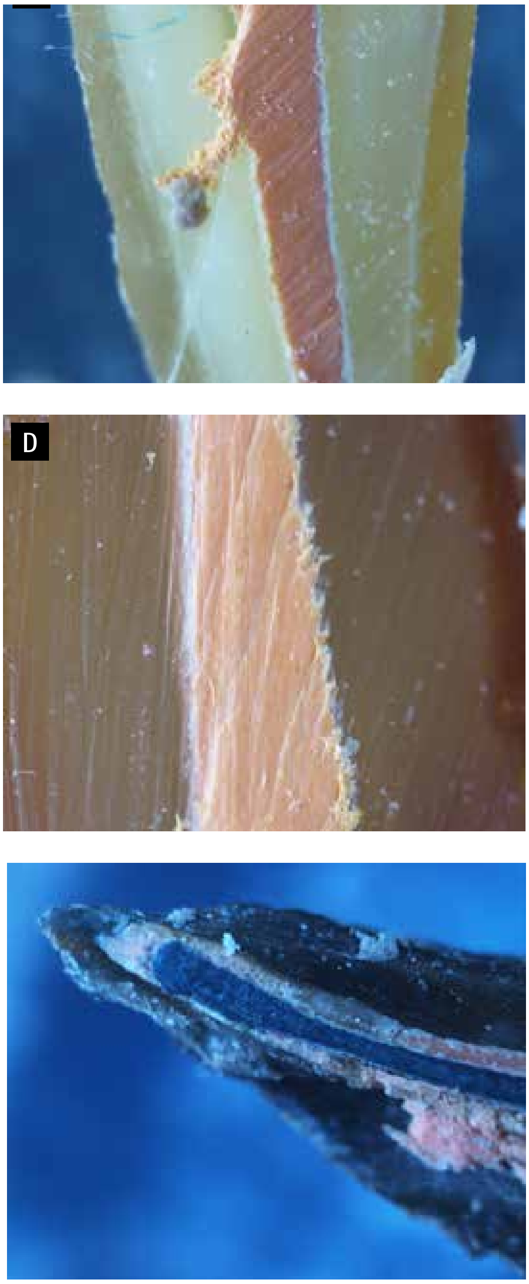


FIG. 2 Group 2 reached an average of 0.665 mm leakage.



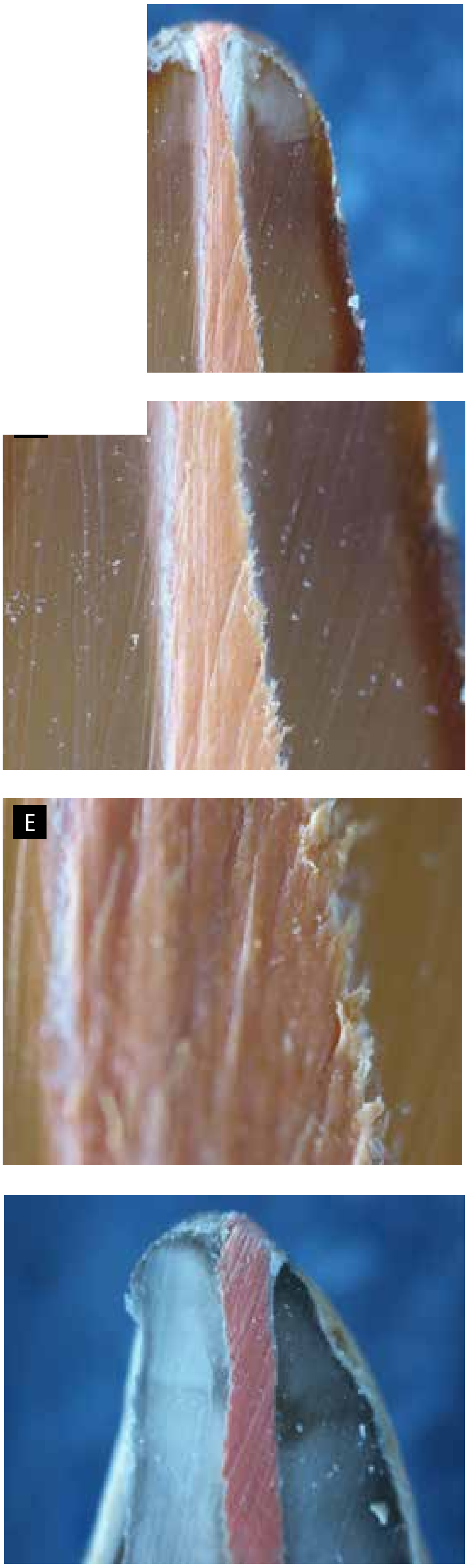


FIG. 4 The highest results belong to Group 1.

FIG. 5 The second highest results belong to Group 2.

endodontic sealers.
2
a
c

A
C
V
a
E
p
S
a
t

D
T
T
I
I

D
T
f
a
p

A
T
s

R

1. Bakland LK, editors. Endodontics. 5th ed. Hamilton, Ontario: BC Decker, 2002. P.571-668,
2. Mannocci F, Innocenti M, Bertelli E, Ferrari M. Dye leakage and SEM study of roots obturated with Thermafill and dentin bonding agent. Endod Dent Traumatol 1999; 15: 60-4.
3. Johnson WB. A new gutta-percha technique. J Endod 1978; 4: 184-8.
4. Becker TA, Donnely JC. Thermafil obturation: a literature review. Gen Dent, 1997; 45: 46-55.
5. Silver GK, Love RM, Purton DG. Comparison of two vertical condensation obturation techniques: touch 'n heat modified and System B. Int Endod J 1999; 32: 287–95.
6. Venturi M, Pasquantonio G, Falconi M, Breschi L. Temperature change within gutta-percha induced by the System-B Heat Source. International Endodontic Journal 2002; 35, 740–6.
7. Viapiana R, Baluci CA, Tanomaru-Filho M, Camilleri J. Investigation of chemical changes in sealers during application of the warm vertical compaction technique. Int Endod J 2015; 48: 16–27.
8. Johnson W, Kulid JC, Tay F. Obturation of the cleaned and shaped root canal system. In: Hargreaves KM, Berman LH, eds. Cohen's Pathways of the Pulp Expert Consult, 11th edn. St. Louis, MO: Mosby, 2016. p 280–323.
9. Viapiana R, Guerreiro-Tanomaru JM, Tanomaru-Filho M, Camilleri J. Investigation of the effect of sealer used on the heat generated at the external root surface during root canal obturation using warm vertical compaction technique with System B heat source. J Endod 2014; 40: 555–61.
10. Friedman S, Torneck CD, Komorowski R, Ouzounian Z, Syrtash P, Kaufman A. In vivo model for assessing the functional efficacy of endodontic filling materials and techniques. J Endod 1997; 23: 557-61.
11. De Moor RJ, Hommez GM. The long-term sealing ability of an epoxy resin root canal sealer used with five gutta percha obturation techniques. Int Endod J 2002; 35: 275-82.
12. Gordon MP, Love RM, Chandler NP. An evaluation of .06 tapered gutta-percha cones for filling of .06 taper prepared curved root canals. Int Endod J 2005; 38: 87-96.
13. Smith CS, Setchell DJ, Harty FJ. Factors influencing the success of conventional root canal therapy--a five-year retrospective study. Int Endod J 1993; 26: 321-33.
14. Schilder H. Filling the root canal in three dimensions. Dent Clin North Am 1967; 11: 723-44.
15. Grossman LI. Endodontic Practice, 10th edn. Philadelphia: Lea & Febiger, 1982.
16. Krell KF, Wefel JS. A calcium phosphate cement root canal sealer--scanning electron microscopic analysis. J Endod 1984; 10: 571-6.
17. Camps J, Jeanneau C, El Ayachi I, Laurent P, About I. Bioactivity of a calcium silicate-based endodontic cement (BioRoot RCS): interactions with human periodontal ligament cells in vitro. J Endod 2015; 41: 1469–73.
18. Dimitrova-Nakov S, Uzunoglu E, Ardila-Osorio H. In vitro bioactivity of Bioroot™ RCS, via A4 mouse pulpal stem cells, has been demonstrated. Dent Mat 2015; 31: 1290–7.
19. Dimitrova-Nakov S, Uzunoglu E, Ardila-Osorio H. The effect of the final irrigant on the antimicrobial activity of root canal sealers. J Dent 2015; 52: 30–6.
20. Arias-Moliz MT, Camilleri J. The effect of the final irrigant on the antimicrobial activity of root canal sealers. J Dent 2016; 52: 30–6.
21. Torabinejad M, Parirokh M, Dummer PMH. Mineral trioxide aggregate and other bioactive endodontic cements: an updated overview. Part II: other clinical applications and complications. Int Endod J 2018 Mar;51(3):284-317.
22. Schwartz RS. Adhesive dentistry and endodontics. Part 2: bonding in the root canal system—the promise and the problems: a review. J Endod 2006; 32: 1125–1134.
23. Kidd EA. Microleakage: a review. J Dent 1976; 4: 199-206.
24. Moccia E, Carpegna G, Dell'Acqua A, Alovisi M, Comba A, Pasqualini D, Berutti, E. Evaluation of the root canal tridimensional filling with warm vertical condensation, carrier-based technique and single cone with bioceramic sealer: a micro-CT study. G Ital Endod 2020; 34:10-7.
25. Limkangwalmongkol S, Burtscher P, Abbott PV, Sandler AB, Bishop BM. A comparative study of the apical leakage of four root canal sealers and laterally condensed gutta-percha. J Endodon 1991; 10: 495-499.
26. Prati C, Gandolfi MG. Calcium silicate bioactive cements: Biological perspectives and clinical applications. Dent Mater 2015; 31: 351-70.
27. Tay FR, Pashley DH. Guided tissue remineralization of partially demineralized human dentine. Biomaterials 2008; 29: 1127-37.
28. Yamamoto S, Han L, Noiri Y, Okiji T. Evaluation of the Ca ion release, pH and surface apatite formation of a prototype tricalcium silicate cement. Int Endod J 2017;50 (Suppl 2: e73-e82). doi: 10.1111/iej.12737..
29. Trope M, Bunes A, Debelian G. Root filling materials and tecniques: biocermics a new hope? Endodontics Topics 2015; 32: 86-96.
30. Gade VJ, Belsare LD, Patil S, Bhede R, Gade JR. Evaluation of push-out bond strength of endosequence BC sealer with lateral condensation and thermoplasticized technique: An in vitro study. J Conserv Dent 2015; 18: 124-7.
31. Celikten B, Uzuntas CI, Orhan A et al. Micro-CT assessment of the sealing ability of three root canal filling techniques. J Oral Sci 2015; 57: 361–6.
32. Camilleri J. Sealers and warm gutta-percha obturation techniques. J Endod 2015; 41: 72-8.
33. DeLong C, He J, Woodmansey KF. The effect of obturation technique on the push-out bond strength of calcium silicate sealers. J Endod 2015; 41: 385-8.
34. Gade VJ, Belsare LD, Patil S, Bhede R, Gade JR. Evaluation of push-out bond strength of endosequence BC sealer with lateral condensation and thermoplasticized technique: An in vitro study. J Conserv Dent 2015; 18: 124-7.
35. Peng L, Ye L, Tan H, Zhou X. Outcome of root canal obturation by warm gutta-percha versus cold lateral condensation: a meta-analysis. J Endod 2007; 33: 106-9.
36. Atmeh AR, Hadis M, Camilleri J. Real-time chemical analysis of root filling materials with heating: guidelines for safe temperature levels. Int Endod J 2020; 53: 698-708.

37. Heran J, Khalid S, Albaaj, Tomson PL, Camileri J. The single cone obturation technique with a modified warm filler. J Dent 2019; 89.
38. Asawaworarit W, Pinyosopon T, Kijsamanmith. Comparison of apical sealing ability of bioceramic sealer and epoxy resin-based sealer using the fluid filtration technique and scanning electron microscopy. J Dent Sci 2020;15:186-192.
39. Yamauchi S, Watanabe S, Okiji T. Effects of heating on the physical properties of premixed calcium silicate-based root canal sealers. J Oral Sci 2020;63:65-69.
40. Chen B, Haapasalo M, Mobuchon C, Li X, Ma J, Shen Y. Cytotoxicity and the effect of temperature on physical properties and chemical composition of a new calcium silicate-based root cana sealer. J Endodont 2020;46:531-8.
41. Akesl H, Makowka S, Bosaid F, Sarkar D, Azim AA. Effect of heat application on the physical properties and chemical structure of calcium silicate-based sealers. Clin Oral Investig 2021; 25: 2717-2725
42. Donnermeyer D, Ibing M, Burklein S, Weber I, Reitze MP, Schafer E. Physico-chemical investigation of endodontic sealers exposed to simulated intracanal heat application: hydraulic calcium silicate-based sealers. Materials (Basel) 2021; 14: 728.
43. Atmeh AR, AlShwaimi E. The effect of heating time and temperature on epoxy resin and calcium silicate-based endodontic sealers. J Endod 2017; 43: 2112–8.
44. De Angelis F, D'Arcangelo C, Buonvivere M, Argentino R, Vadini M. In vitro leakage evaluation of bioceramic and zinc-eugenol sealers with two obturation techniques. Coatings 2021; doi.org/10.3390/coatings11060727
45. Jeong JW, DeGraft-Johnson A, Dorn SO, Di Fiore PM. Dentinal Tubule penetration of a calcium silicate–based root canal sealer with different obturation. J Endod 2017; 5: 1–5.